

- 9 A particle moves along the curve $y = \frac{3(x+6)}{x+4}$, where $x \neq -4$, in such a way that the y -coordinate of the particle is increasing at a constant rate of $\frac{4}{27}$ units per second.

Find the x -coordinates of the particle at that instant that the x -coordinates of the particle is decreasing at a rate of 2 units per second.

[5]